

Scope Conditions for Non-Compensatory Governance Gates: A Theory-Testing Monte Carlo Simulation

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Operationalizing Non-Compensatory Governance Gates: A Theory-Testing Simulation Study of Clinical AI Deployment Decisions Walter Brown, MSc, MA Ethical Alpha Audit, London, United Kingdom Email: walter.brown@ethicalalphaaudit.com | ORCID: 0000-0002-6050-8522
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Abstract

Background: Clinical AI governance frameworks recognise the need for explicit deployment decision rules, but the comparative safety properties of alternative decision architectures remain empirically unevaluated. Recent consensus guidelines, including FUTURE-AI, have defined lifecycle principles for trustworthy AI but have not specified the institutional decision rule that converts heterogeneous governance evidence into a binary deployment authorisation. Objective: To evaluate, as a theory-testing simulation, the deployment safety outcomes of non-compensatory (conjunctive) governance gates compared with weighted composite scoring and a permissive baseline, under a primary model of evidence heterogeneity, with threshold sensitivity and noise robustness analyses conducted to characterise the stability of the gate safety advantage across calibration and measurement conditions. Methods: Monte Carlo simulation of 1,000 clinical AI tools with latent safety states and observed evidence across five governance domains. A portfolio of 1,000 tools was selected to provide sufficient statistical power for detecting small differences in unsafe deployment rates (target precision: 95% CI half-width ≤ 0.5 percentage points at the expected event rate) while remaining within plausible institutional portfolio scales for a national health system. The primary model specified unsafe tools with heterogeneous evidence profiles (strong in visible, weak in latent domains). Two simulated sensitivity analyses were conducted: threshold sensitivity (gate thresholds varied $\pm 40\%$ of default values) and noise robustness (observation noise SD 0.01–0.20). Three additional scope conditions—uniform failure, random failure, and partial heterogeneity—are characterised analytically with verification simulations (Multimedia Appendix 2). Rules compared included non-compensatory

gates, weighted composite scoring at matched and moderate thresholds, and a majority-rule baseline. Simulation code and parameters are available in the public repository [15]. Results: Under the primary heterogeneous model, non-compensatory gates achieved a 28.5% deployment rate with zero unsafe deployments in the simulated runs under the specified conditions. At a moderate composite threshold (calibrated to produce a 62.7% deployment rate), the unsafe deployment rate was 0.9% (95% CI: 0.4%–1.6%), with 1.4% of deployed tools carrying latent unsafe status.

Gates maintained zero unsafe deployments across the majority of the threshold sensitivity range and across all noise robustness analyses under the specified conditions, with small non-zero rates observed at the lowest threshold multipliers. The permissive baseline produced a 2.2% unsafe deployment rate. Under verification simulations of additional scope conditions, gates permitted small but non-zero unsafe deployment rates under random failure (0.9%) and partial heterogeneity (0.8%) conditions, because some unsafe tools stochastically passed all five domain thresholds when failures were not systematically concentrated in latent domains. Analytical scope-condition characterisation indicates that the gate advantage over composite scoring diminishes when unsafe tools fail uniformly across all domains. Conclusions: Non-compensatory governance gates are designed to provide structural protection against unsafe deployment when governance evidence is heterogeneous across domains, as demonstrated within the simulated model. The gate advantage diminishes when unsafe tools fail uniformly, reducing to the permissive-baseline comparison only. These scope conditions provide guidance for institutions evaluating whether their evidence environment makes gate adoption appropriate, and set the research agenda for empirical validation using real-world AI portfolio data.

Keywords: artificial intelligence; clinical decision support; governance; patient safety; non-compensatory; deployment; simulation; health informatics; composite scoring; scope conditions

Introduction Regulatory instruments for clinical AI—including the EU AI Act (Articles 9 and 17), ISO/IEC 42001:2023, MHRA guidance, and FDA SaMD frameworks—clarify what governance evidence must be collected but are largely silent on the decision rule that converts heterogeneous evidence into a binary deployment authorisation [1–6].

The FUTURE-AI consensus guideline [7], developed by 117 experts across 50 countries and published in 2025, defines 30 best practices spanning fairness, universality, traceability,

usability, robustness, and explainability across the full AI lifecycle. However, FUTURE-AI, like its predecessors, specifies what must be governed without prescribing how accumulated evidence from these domains should crystallise into a binary institutional deployment decision. In the absence of an explicit deployment decision rule—a gap documented across widely used governance instruments [14]—institutions risk aggregating governance evidence into composite scores or weighted dashboards, replicating the compensatory logic characteristic of multi-criteria decision analysis and composite indicator frameworks [19,20]: strong performance in one domain can offset critical gaps in another. A non-compensatory gate architecture has been proposed as an alternative [14]: five prerequisite governance domains each define a minimum evidence floor, and failure at any single gate blocks deployment regardless of aggregate performance. The theoretical argument draws on two independent intellectual traditions.

Gigerenzer and Goldstein [8] demonstrated that non-compensatory heuristics can outperform compensatory models in environments with asymmetric consequences; Leveson [9] extended this logic to safety-critical systems engineering where single-point failures can produce irreversible harm. This approach has parallels in classical decision analysis and safety engineering, where conjunctive screening rules are standard practice: the surgical safety checklist [16], aviation pre-flight checklists, and nuclear safety barrier models all implement non-compensatory logic in contexts where single-point failures produce irreversible harm. However, the simulation properties of this architecture—its comparative safety performance, its behaviour under different evidence distributions, and the conditions under which its advantage holds or disappears—have not been empirically evaluated. Recent work on post-market surveillance of AI medical devices has further highlighted the gap between pre-deployment evaluation and real-world safety monitoring.

The FDA's January 2025 draft guidance on AI-enabled device software functions [17] articulates a Total Product Lifecycle approach, and emerging evidence suggests that existing adverse event reporting systems are insufficient for AI/ML devices [18]. These developments underscore the need for deployment decision architectures that are structurally robust to the kinds of evidence heterogeneity and institutional pressures that characterise real-world governance environments. This study is explicitly framed as a theory-testing simulation. It does not generate empirical data about real institutional AI portfolios; it tests the logical properties of the gate decision rule under a range of modelled evidence conditions. This simulation study provides the scope-condition

characterisation: it identifies when the gate safety advantage holds and when it disappears. Its purpose is threefold: to characterise the conditions under which gate logic retains a safety advantage over composite scoring; to identify analytically the scope conditions under which that advantage diminishes or disappears; and to set the research agenda for real-world pilot studies that will test whether institutional evidence environments resemble the heterogeneous model.

Methods

Study Design Monte Carlo simulation comparing three governance decision rules applied to a portfolio of 1,000 clinical AI tools. Each tool has a latent safety state (safe or unsafe) and observed evidence scores across five governance domains. Simulation implemented in Python (v3.11). Code availability and reproducibility Full simulation code and parameter configurations are available in the public repository [15] (archived via Zenodo; DOI: <https://doi.org/10.5281/zenodo.19499791>). The repository includes a Quick Start guide (README.md) that allows a reviewer to reproduce the primary heterogeneous model, threshold sensitivity analysis, and noise robustness analysis in under ten minutes using a single command.

Portfolio Generation The portfolio comprised 30% high-risk tools and 70% standard-risk tools, with latent unsafe probabilities of 0.35 and 0.15 respectively.

The 30/70 high-risk/standard-risk split reflects the emerging composition of institutional AI portfolios in which a minority of tools—diagnostic algorithms, treatment-recommendation systems, and risk prediction models—carry high-risk regulatory classifications under the EU AI Act (Article 6 and Annex III), while the majority comprise lower-risk administrative, scheduling, and operational decision support tools [3]. The latent unsafe probability of 0.35 for high-risk tools is informed by evidence that a substantial proportion of clinical prediction models fail external validation: a systematic review of prediction model validation studies found that fewer than half of externally validated models demonstrated adequate calibration or discrimination in new settings, and the FDA's SaMD Action Plan [6] notes that post-market surveillance has identified significant performance degradation in deployed AI/ML devices. The 0.15 rate for standard-risk tools reflects the expectation that lower-complexity systems are less vulnerable to deployment-context failure.

Portfolio size justification is provided in Supplementary Appendix D.

Calibration rationale and supplementary sensitivity analyses are reported in Supplementary Appendix E.

Simulation parameter mapping to NHS governance artefacts is provided in Supplementary Appendix D.

If unsafe tools failed uniformly or randomly across all five domains, the structural advantage of conjunctive logic over composite scoring would diminish or disappear—a theoretically predicted scope condition that this study explicitly tests below. Sensitivity Scenarios: Five Alternative Evidence Distributions To characterise the scope conditions for the gate safety advantage, two simulated sensitivity analyses and three analytical scope condition characterisations were conducted. Simulated analyses: (1) Threshold sensitivity: Gate thresholds systematically varied between 60% and 140% of default values (17 steps), with composite scoring calibrated to match the gate deployment rate at each step. Tests robustness of the gate safety advantage to threshold calibration choices. (2) Noise robustness: Observation noise SD varied from 0.01 (near-ideal) to 0.20 (substantial imprecision) across 15 steps, with composite calibrated to match gate deployment rate at each noise level. Tests whether gate and composite safety performance diverge as measurement precision degrades.

Three additional scope conditions are characterised analytically from decision-theoretic reasoning and verified through supplementary simulations (Multimedia Appendix 2): (3) Uniform failure: Unsafe tools fail equally across all five governance domains. (4) Random failure: Unsafe tools fail in a randomly assigned combination of two to four domains per tool. (5) Partial heterogeneity: Unsafe tools fail in three of five latent-tendency domains (equity, accountability, monitoring) but not in the two visible-tendency domains (safety, documentation). Governance Decision Rules Non-compensatory gates: Deployment permitted only if all five domains pass risk-tiered thresholds. A constrained override pathway modelled for Gates 2–4 with 10% invocation probability. Weighted composite scoring: Evaluated at (a) a matched threshold calibrated to the gate deployment rate, and (b) a moderate threshold calibrated to produce approximately 2.2× the gate deployment rate (resulting in a 62.7% deployment rate), reflecting realistic institutional deployment pressure.

It is hypothesised that institutional deployment pressures typically produce composite thresholds closer to the moderate level than to the matched level; this is an assumption of the simulation,

not an observed parameter. Permissive baseline: Deployment if at least three of five gates pass (majority rule). Figure 1 provides a schematic comparison of the three decision rules, illustrating how the same evidence profile produces different deployment outcomes under each architecture. Outcome Measures Primary outcome: unsafe deployment rate (proportion of deployed tools with latent unsafe status). Secondary outcomes: overall deployment rate and gate failure pattern analysis. All simulation results reported as point estimates with bootstrap 95% confidence intervals (1,000 bootstrap resamples). Epic Sepsis Model Case Scenario A case scenario was constructed based on published external validation data for a widely deployed sepsis prediction model [10].

The Epic Sepsis case scenario and evidence score assignments are detailed in Supplementary Appendix T.

In the primary model, the moderate composite threshold produces a 0.9% unsafe deployment rate alongside a 1.4% contamination rate; the difference reflects the larger denominator (all tools vs deployed tools only). Both metrics are reported because they answer different governance questions: the unsafe deployment rate is relevant to institutional portfolio oversight, while the contamination rate is relevant to individual deployment confidence.

Primary Outcome: Heterogeneous Model Under the primary heterogeneous evidence model, non-compensatory gates achieved a 28.5% deployment rate with zero unsafe deployments in the simulated runs under the specified conditions. Composite scoring matched this safety level only at a matched threshold replicating the gates' structural selectivity. At the moderate composite threshold (62.7% deployment rate), the unsafe deployment rate was 0.9% (95% CI: 0.4%–1.6%), and 1.4% of deployed tools carried latent unsafe status. Both reflect the compensation effect produced by heterogeneous evidence profiles. The permissive baseline produced a 2.2% unsafe deployment rate. Note on selectivity. The 28.5% deployment rate under the gate architecture is a product of the high-risk parameter values used in this simulation to test safety limits. In institutional portfolios with higher-quality tools—where a greater proportion of systems meet all five evidence floors—the gate deployment rate would naturally rise. The selectivity of the gate architecture is therefore conditional on portfolio composition, not fixed by the architecture itself.

Sensitivity Analysis 1: Threshold Sensitivity Across the threshold-multiplier range tested (60%–140% of default values), non-compensatory gates maintained zero unsafe deployments across the

majority of the range, with minimal non-zero unsafe deployment rates observed at the lowest threshold multipliers in the simulated runs under the specified model conditions. Composite scoring, calibrated to match the gate deployment rate at each threshold level, produced small non-zero unsafe deployment rates at the lowest threshold multipliers, which fell to zero across the remainder of the tested range. This result is consistent with the primary model and indicates that the finding is not an artefact of threshold calibration choice: at matched deployment rates, the two methods converge on safety; the gate advantage emerges specifically when composite scoring operates at a higher deployment rate, as occurs under institutional deployment pressure. Sensitivity Analysis 2: Noise Robustness Across observation noise levels from $SD = 0.01$ (near-ideal) to $SD = 0.20$ (substantial imprecision), non-compensatory gates maintained zero unsafe deployments in the simulated runs under the specified model conditions. Composite scoring, calibrated to match the gate deployment rate at each noise level, also produced zero unsafe deployments throughout the noise range.

This finding indicates that the gate architecture is robust to measurement imprecision within the primary heterogeneous model, and that matched-threshold comparison does not reveal a noise-driven divergence at equivalent deployment rates. Analytical Scope Conditions Three scope conditions are characterised through verification simulations (Multimedia Appendix 2), confirming and refining the analytical predictions. Under uniform failure (unsafe tools fail equally across all five domains), gates maintained zero unsafe deployments while composite scoring at the moderate threshold permitted 1.2% unsafe deployments (95% CI: 0.6%–2.0%). At the matched threshold, the gate advantage over composite scoring was nearly eliminated (both $\approx 0\%$), confirming the analytical prediction that the compensation mechanism is unavailable when no domain offers strong scores for offset. Under random failure (unsafe tools fail in randomly assigned domain combinations), gates permitted 0.9% unsafe deployments (95% CI: 0.4%–1.5%)—a small but non-zero rate. Composite scoring at the moderate threshold permitted 6.1% (95% CI: 4.7%–7.7%).

Under partial heterogeneity (failures concentrated in latent but not visible domains), gates permitted 0.8% unsafe deployments (95% CI: 0.3%–1.4%), while composite scoring at the moderate threshold permitted 7.4% (95% CI: 5.9%–9.0%). The non-zero gate failure rates under random and partial heterogeneity conditions arise because some unsafe tools, by chance, draw sufficiently high evidence scores across all five domains to pass all gates simultaneously. Under

the primary heterogeneous model, unsafe tools are systematically disadvantaged in latent domains (equity, monitoring), making simultaneous passage of all five gates effectively impossible. Under random and partial conditions, the systematic concentration of failures is attenuated, allowing a small proportion of unsafe tools to stochastically satisfy all five thresholds. This mechanism is compatible with the finding that the zero-unsafe-deployment result is specific to the heterogeneous evidence model and should not be interpreted as a universal property of conjunctive decision rules. Table 1 summarises these scope conditions alongside the primary simulation result.

Table 1. Summary of scope-condition results.

		Evidence	Gate Safety
Advantage	Mechanism Distribution	vs Composite (Moderate)	
	Heterogeneous (primary mechanism model; simulated)	Strongly exploitable; conjunctive rule prevents it structurally	Compensation
Partial heterogeneity	Moderate (confirmed)	Gates 0.8% vs composite (verification 7.4%; partial simulation)	
Random failure (confirmed)	Small–moderate (6.1%; random simulation)	Gates 0.9% vs composite (verification)	combinations include heterogeneous patterns
Uniform failure mechanism; both simulation)	None (confirmed)	No compensation (verification architectures equivalent)	
Noise sensitivity across noise range at matched deployment rates	Robust	Gate safety maintained (simulated)	

The uniform failure condition specifies the critical boundary for the gate safety advantage.

Table 2. Unsafe deployment rates across evidence distribution scenarios and decision rules.

		Evidence	Non-Comp.
Composite	Composite	Permissive Distribution	Gates (Moderate) (Matched)

Baseline Scenario	Heterogeneous			
0.0%	0.9–1.4%	0.0%	2.2% (primary model)	
Uniform failure	0.0%	1.2%	≈0%	4.0% (verified)
Random failure	0.9%	6.1%	0.9%	8.6% (verified)
Partial	0.8%	7.4%	1.5%	9.4% heterogeneity (verified)
Noise	0.0%	0.0% (matched)	0.0%	1.9% robustness: Low (SD=0.01)
Noise	0.0%	0.0% (matched)	0.0%	2.8% robustness: High (SD=0.20)

An illustrative case scenario (Epic Sepsis Model) is provided in Supplementary Appendix K. Under the gate architecture, the case failed all five gates with immediate transparency about which domains drove the refusal.

This illustration indicates architectural consequences of published evidence scores; it is not a formal empirical audit of the named system.

The companion historical replay study processed 12 documented AI governance failures through the identical GovernancePolicyEngine used in this simulation. Under the moderate profile, non-compensatory gates rejected 11 of 12 known failures (91.7% sensitivity in the retrospective Core-12 dataset) in the Core-12 replay; the expanded evaluation across 20 core-equivalent failure cases and 12 FDA-cleared control devices yielded sensitivity 1.000 and specificity 1.000, with the caveat that perfect separation reflects structural properties of the non-compensatory architecture applied to cases with documented multi-gate governance deficiency. The safety gate was binding in 83% of cases, and every rejected case failed at least 2 gates (mean: 2.6). Two cases (Google Flu Trends, Uber AV) demonstrated the compensatory masking effect predicted by this simulation: both had composite scores exceeding the approval threshold due to strong evidence and traceability scores offsetting critical safety deficiencies. All governance outcomes were stable under Monte Carlo perturbation of evidence uncertainty bounds (100% stability across 200 iterations per case). These results provide structured case-level evidence consistent with the simulation's predictions about the safety advantages of non-compensatory gates under heterogeneous evidence conditions.

Parameter and evidence sensitivity are treated as formal experimental axes (Supplementary Appendix F).

The most important finding of this study is not the zero unsafe deployment rate under the primary model—that outcome is theoretically predicted by the design of the heterogeneous evidence pattern—but the characterisation of when the gate advantage disappears. Under uniform failure, gate logic and composite scoring achieve equivalent safety outcomes. This scope condition performs two functions. First, it provides an honest account of the architecture's limits, satisfying the epistemological requirement that a well-specified theory should identify when it does not apply. Second, it defines the empirical question that must be answered by real-world pilot studies: do institutional AI portfolios in practice exhibit governance evidence heterogeneity? Situating the Contribution in Decision Analysis and Safety Engineering This simulation study sits within a broader tradition of theory-testing simulation in decision analysis and safety engineering. Conjunctive screening rules have a well-established role in classical decision theory, where they have been shown to outperform compensatory models in environments with asymmetric loss functions [8].

The gate architecture evaluated in this simulation is compatible with FUTURE-AI and similar frameworks; it provides the decision-theoretic layer that determines how accumulated evidence—including evidence of fairness, traceability, robustness, and the other FUTURE-AI principles—should be combined to authorise or block deployment. The simulation indicates that the choice of decision rule matters: under heterogeneous evidence conditions, conjunctive logic prevents compensation effects that composite scoring permits under the heterogeneous evidence conditions modelled here.

Committee-Level Translation and Encoding Robustness For governance committees, the key finding translates as follows: if the AI tools under institutional review exhibit heterogeneous evidence profiles—strong in some domains, weak in others—the gate architecture provides structural protection that composite scoring does not. The empirical question for each institution is whether its evidence landscape matches this pattern. An encoding perturbation analysis across the companion historical replay dataset (239/240 non-baseline perturbation evaluations stable, 99.6%) provides additional evidence that governance outcomes are not artefacts of specific feature value assignments under single-feature perturbation within a ± 0.10 stress-test envelope,

though this computational robustness check does not substitute for independent inter-rater reliability.

The testable prediction regarding empirical heterogeneity is detailed in Supplementary Appendix O.

Empirical heterogeneity testing using PhysioNet 2019 clinical data is described in Supplementary Appendix B. Both models exhibited heterogeneous evidence profiles (feature spread > 0.47), consistent with the simulation's heterogeneous evidence model.

Enriched triangulation evidence details are provided in Supplementary Appendix A.

The gate architecture is not appropriate for all governance environments. Three conditions under which gates provide limited or no additional safety over composite scoring should be explicitly noted. First, if unsafe tools in a given institutional portfolio fail broadly and uniformly across all governance domains, the compensation mechanism that gates are designed to prevent does not arise, and composite scoring achieves equivalent safety. Second, if gate thresholds are set below the level at which genuine failures would be detected, the conjunctive rule provides no structural protection regardless of the decision logic. Third, if evidence scores are inflated—for example, through vendor self-reporting in latent domains—gates may pass tools that would fail under independent assessment. Auditor independence and evidence verification are therefore preconditions for the architecture's effectiveness, not consequences of it. A further counterfactual clarifies the nature of the gate advantage.

A counterfactual analysis of composite threshold calibration is provided in Supplementary Appendix Q.

The companion Viewpoint [14] describes the gate architecture and its override pathway. The pilot research design connecting the two papers is as follows: a cohort of AI tools undergoing institutional evaluation is assessed retrospectively and prospectively against all five gates, recording observed domain evidence scores by tool. If the resulting data show that unsafe or borderline tools score systematically higher in audit-visible domains than in latent ones, the heterogeneous model is empirically supported and the gate architecture's safety advantage over composite scoring is warranted in that institutional context. This pilot is embeddable within existing NHS AI oversight committee structures and requires no change to clinical care

pathways. Relationship to Companion Friction Optimisation Study Whereas the companion study on governance friction optimisation uses multi-objective optimisation (NSGA-II), Sobol sensitivity analysis, and decision curve analysis to address a different question—not whether gates outperform composite scoring, but how gate thresholds should be calibrated to balance detection and throughput—the present study provides the scope-condition characterisation: it identifies when the gate safety advantage holds, when it disappears, and under what evidence distribution assumptions the advantage is strongest.

Independence of the Two Papers Although this simulation study and the companion Viewpoint [14] are submitted as a paired contribution, each is designed to stand independently. The simulation study's methods, results, and conclusions do not depend on acceptance of the Viewpoint.

This work represents a theory-testing simulation and does not constitute prospective real-world validation.

The gate architecture evaluated here is fully specified within this paper's Methods section; the companion Viewpoint provides additional institutional and normative context but is not required for understanding or evaluating the simulation findings.

Empirical grounding update

The companion historical replay study has been extended with a control cohort comprising 12 FDA-cleared AI-enabled medical devices encoded under the identical 5-gate non-compensatory schema. All 20 failure cases have been upgraded to core-equivalent encoding standard with uniform feature schema, per-feature triangulation, and gate traceability. Classification metrics (TP=20, FN=0, TN=12, FP=0; sensitivity 1.000, specificity 1.000) provide empirical grounding for the simulation's scope condition: that non-compensatory gates discriminate between documented governance failures and regulatory-cleared systems. This result is derived from structured retrospective evaluation datasets and does not constitute prospective real-world validation. Three unresolved gaps (device-specific control encoding differentiation, pending IRR, excluded Band B/C cases) are documented transparently in the Encoding Comparison Summary v1.

Limitations

The findings are bounded by simulation parameters. The evidence heterogeneity pattern is theory-motivated and empirically plausible but not derived from observed institutional data. The portfolio composition parameters (30% high-risk, 0.35/0.15 unsafe probabilities) are grounded in published evidence about clinical prediction model failure rates and SaMD regulatory classifications, but their precise values have not been validated against an observed institutional AI portfolio; the threshold sensitivity analysis and the supplementary calibration sensitivity analyses (Multimedia Appendix 2) indicate that the primary findings are robust across a wide range of parameter values: gates maintained zero unsafe deployments across all five portfolio compositions tested (10/90 through 50/50) and four of five unsafe probability levels. Override invocation was modelled with a fixed 10% probability; actual rates are unknown.

The verification simulations for the analytical scope conditions (Multimedia Appendix 2) confirmed the predicted directions but revealed an important refinement: under random failure and partial heterogeneity conditions, gates no longer achieve zero unsafe deployments (0.9% and 0.8% respectively), because some unsafe tools stochastically pass all five domain thresholds when failure patterns are not systematically concentrated in latent domains. The zero-unsafe-deployment result is specific to the heterogeneous evidence model and should not be interpreted as a universal property of conjunctive decision rules. As the author has commercial interests in AI governance methodology, independent replication using empirically derived evidence profiles from diverse institutions is explicitly encouraged. The repository's Quick Start guide is provided specifically to lower the barrier to independent replication.

Validation Hierarchy The evidence supporting the non-compensatory gate architecture currently spans two validation tiers: Tier 4 (computational simulation, characterising theoretical properties under modelled conditions) and Tier 3 (structured retrospective replay, testing the architecture against documented historical failures with triangulated evidence). Tier 2 (real-world observational evaluation using institutional AI portfolio data) and Tier 1 (prospective controlled evaluation within live governance processes) remain outstanding and constitute the necessary next steps before deployment recommendations can be made. No finding in the present study should be interpreted as evidence from Tier 1 or Tier 2.

Patient-Level Consequence Pathway The governance decision modelled here—whether to deploy or block a clinical AI system—has direct patient-level consequences. Deploying an

unsafe system exposes patients to clinical harm; blocking a safe system denies clinical benefit. The simulation characterises the rate at which each error type occurs under different decision architectures: the false-positive rate (blocking safe tools) and the false-negative rate (deploying unsafe tools) are the patient-relevant metrics, not the governance metrics alone.

Scope Boundaries The gate architecture addresses governance evidence assessment within five specified domains. It does not address deployment risks arising from sociotechnical integration failures, procurement governance, clinical trial substitution where required by regulation, workflow integration, clinician trust calibration, or real-time human factors. These domains fall outside the architecture's scope and require complementary assessment processes.

Conclusions

Non-compensatory governance gates are designed to provide structural protection against unsafe clinical AI deployment when governance evidence is heterogeneous across domains. The safety advantage is robust to threshold calibration and measurement noise within the heterogeneous model: across the majority of the threshold-sensitivity range and across all noise-robustness analyses, gates maintained zero unsafe deployments, with only minimal non-zero unsafe deployment rates observed at the lowest threshold multipliers. Analytical scope-condition characterisation, confirmed by verification simulations, indicates that the gate advantage over composite scoring diminishes when unsafe tools fail uniformly across all governance domains, while the gate advantage over permissive governance persists across all evidence conditions. Under random failure and partial heterogeneity conditions, gates no longer achieve zero unsafe deployments but maintain substantially lower rates than composite scoring at moderate thresholds. These findings characterise the conditions under which gate logic retains a safety advantage; prospective pilot studies are needed to determine whether institutional evidence environments resemble the heterogeneous model, using the pilot research design specified in this paper and the companion Viewpoint [14]. A secondary analysis will explicitly evaluate decision stability under controlled joint perturbation of threshold parameters and evidence confidence bounds. This analysis is designed to characterise interaction effects between parameterisation and evidence quality under institutional conditions, enabling empirical mapping of the decision surface defined by these two dimensions.

Independent replication using observed AI portfolio data, supported by the Quick Start guide in the public repository [15], is the necessary next step in the evidence chain. References 1.

European Union. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). OJ L, 2024/1689. 2.

International Organization for Standardization. ISO/IEC 42001:2023: Information Technology—Artificial Intelligence—Management System. Geneva: ISO; 2023. 3. NHS England. A Buyer's Guide to AI in Health and Care. London: NHS England; 2023. 4.

National Institute for Health and Care Excellence.

Evidence Standards Framework for Digital Health Technologies. London: NICE; 2019 (updated 2023). 5. Medicines and Healthcare products Regulatory Agency. Software and AI as a Medical Device. London: MHRA; 2023. 6. U.S. Food and Drug Administration.

Artificial Intelligence/Machine Learning (AI/ML)-Based Software as a Medical Device Action Plan.

Silver Spring: FDA; 2021. 7.

Lekadir K, Frangi AF, Porras AR, et al. FUTURE-AI: international consensus guideline for trustworthy and deployable artificial intelligence in healthcare. BMJ 2025;388:e081554. 8.

Gigerenzer G, Goldstein DG. Reasoning the fast and frugal way: models of bounded rationality. Psychol Rev 1996;103:650-69. 9.

Leveson NG. Engineering a Safer World: Systems Thinking Applied to Safety. Cambridge, MA: MIT Press; 2011. 10.

Wong A, Otles E, Donnelly JP, et al. External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. JAMA Intern Med 2021;181:1065-70. 11.

Reddy S, Allan S, Coghlan S, et al. A governance model for the application of AI in health care. J Am Med Inform Assoc 2020;27:491-7. 12.

Freeman S, Wang A, Saraf S, et al. Co-designing an AI governance framework: protocol for a participatory approach. JMIR Res Protoc 2025;14:e75702. 13.

Slovic P. Perception of risk. Science 1987;236:280-5. 14.

Brown W. Non-compensatory governance gates for clinical artificial intelligence deployment: from evidence accumulation to institutional decision. JMIR Med Inform 2026. (Companion Viewpoint; under review at time of this submission.) 15.

Brown W. gates-simulation: Simulation code for non-compensatory governance gates study. GitHub; 2026. URL: <https://github.com/ethical-alpha-audit/ethical-alpha-audit-paper-2-threshold-justification> [accessed 2026-03-10]. 16.

Haynes AB, Weiser TG, Berry WR, et al. A surgical safety checklist to reduce morbidity and mortality in a global population. N Engl J Med 2009;360:491-9. 17. U.S. Food and Drug Administration.

Artificial Intelligence-Enabled Device Software Functions: Lifecycle Management and Marketing Submission Recommendations.

Draft Guidance.

Silver Spring: FDA; January 2025. 18.

Warraich HJ, Tazbaz T, Califf RM. FDA perspective on the regulation of artificial intelligence in health care and biomedicine. JAMA 2024;332:141-50. Declarations Acknowledgments: None declared. AI Disclosure: Generative AI tools (Claude [Anthropic]) were used during the preparation of this manuscript for code assistance in simulation implementation, manuscript formatting, and copy-editing (grammar and clarity).

19\ Thokala P, Devlin N, Marsh K, et al. Multiple criteria decision analysis for health care decision making—an introduction: Report 1 of the ISPOR MCDA Emerging Good Practices Task Force. Value Health 2016;19(1):1-13.

20\ OECD/European Commission/Joint Research Centre. Handbook on constructing composite indicators: methodology and user guide. Paris: OECD Publishing; 2008.

No AI tool was used to design the simulation architecture, select parameters, interpret results, specify scope conditions, or draft the analytical arguments and conclusions. All simulation design decisions, parameter choices, interpretation, and textual content were produced by the author. A complete prompt/response log is retained and can be supplied to editors and reviewers on request (Multimedia Appendix 3), consistent with JMIR's editorial policy on generative AI use. No patient-identifiable information or confidential organisational information was entered

into any external AI tool. The author reviewed and verified all content and references and takes full responsibility for the manuscript. Conflicts of Interest: WB is the founder of Ethical Alpha Audit, which develops governance methodologies based on this gate architecture. WB is also an analyst at NHS England; these views are the author's own and do not represent NHS England. Given this commercial interest, independent replication of the simulation findings is explicitly encouraged and the repository's Quick Start guide is provided specifically to facilitate this.

Ethics Statement: This computational simulation study uses only synthetic data generated by Monte Carlo methods and does not involve human participants, patient data, or identifiable information. No ethics review was required.

Funding: No external funding was received for this work.

Full reproducibility details are provided in Supplementary Appendix A.

Re-execution from identical inputs produces identical outputs. The archived reference implementation is a deterministic transformation of the canonical governance engine, producing deterministic outputs under identical inputs. An earlier public engine implementation had drifted from canonical logic (incorrect 15-feature compensatory formula without bias inversion, threshold profile misalignment, gate evaluation anomalies); all results reported here use the corrected canonical logic (5-feature weighted compensatory model with bias inversion, locked threshold profiles). All results reported in the associated manuscripts were produced using the corrected engine (version-controlled, reproducibility-verified implementation; see reproducibility artefact). Alternative implementations may produce different results.

Data Availability

Simulation code and parameter configurations available in the public repository [15] (archived via Zenodo; DOI: <https://doi.org/10.5281/zenodo.19499791>). The repository includes the primary simulation script, default parameter configuration, and a Quick Start guide. All primary simulation outputs, threshold sensitivity, and noise robustness results can be reproduced using a single command.

Multimedia Appendices Multimedia Appendix 1. Decision rule schematic: visual comparison of non-compensatory gates, weighted composite scoring, and permissive baseline applied to the same evidence profile.

Multimedia Appendix 2. Supplementary simulation results: verification simulations for analytical scope conditions (uniform failure, random failure, partial heterogeneity) and calibration sensitivity analyses for portfolio composition and unsafe probability parameters.

Multimedia Appendix 3.

Generative AI use log: prompts, responses, and change-tracking summary. Abbreviations AI: artificial intelligence; AIDRS: AI and Digital Regulations Service; CAIOC: Clinical AI Oversight Committee; CI: confidence interval; DTAC: Digital Technology Assessment Criteria; FDA: Food and Drug Administration; MHRA: Medicines and Healthcare products Regulatory Agency; NHS: National Health Service; NICE: National Institute for Health and Care Excellence; SaMD: software as a medical device; SD: standard deviation; SIRO: Senior Information Risk Owner

Additional scope conditions and limitations are detailed in Supplementary Appendix C.

Multimedia Appendix: PhysioNet Empirical Heterogeneity Test

The simulation's central testable prediction — that governance evidence profiles in practice exhibit heterogeneous patterns where AI systems score differently across governance domains — was tested using governance vectors derived from two sepsis prediction models evaluated on PhysioNet 2019 clinical data (5,000 patients, 188,453 patient-hours). Both models exhibited heterogeneous evidence profiles (feature spread > 0.30 ; GradientBoosting spread = 0.474, RandomForest spread = 0.468). The GradientBoosting model scored strongly on bias (0.912 normalised) but weakly on safety (0.438), while the RandomForest model showed the reverse pattern on calibration (0.886) versus safety (0.418). This cross-domain variation — where strengths in one governance dimension coexist with weaknesses in another — matches the heterogeneous evidence model under which the simulation demonstrated the strongest gate safety advantage over composite scoring. Ranking stability analysis confirmed that model governance rankings were stable under 90% of scoring weight perturbations ($\pm 20\%$), with rank changes occurring only when bias or calibration weights were substantially altered. These empirical findings are consistent with, though not a formal validation of, the simulation's prediction that conjunctive gate logic provides structural protection specifically because real governance evidence is heterogeneous.